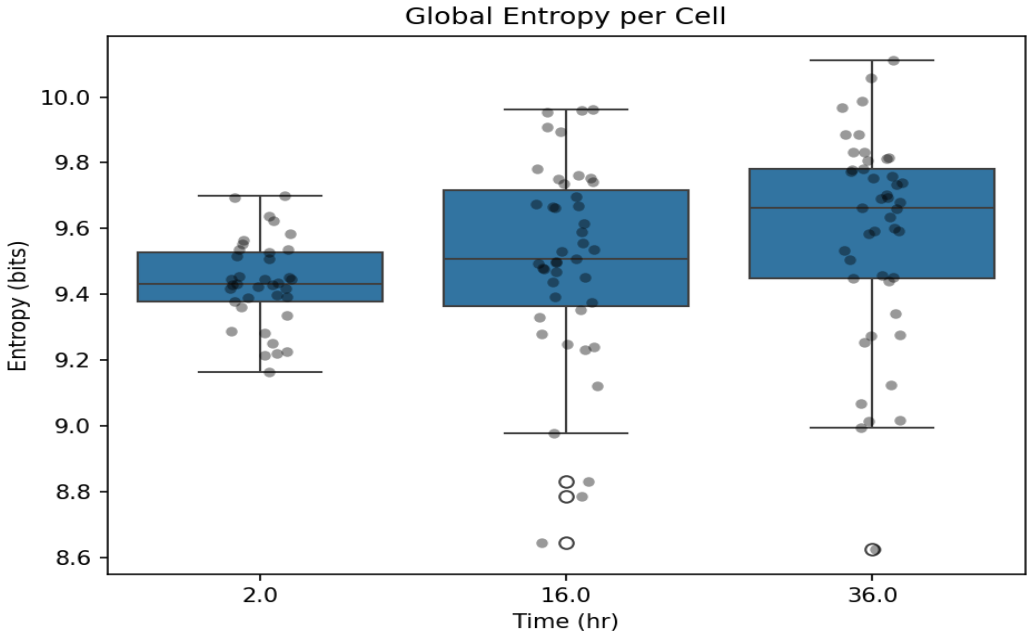
Information■Theoretic Analysis of Yeast scRNA■seq

Methods

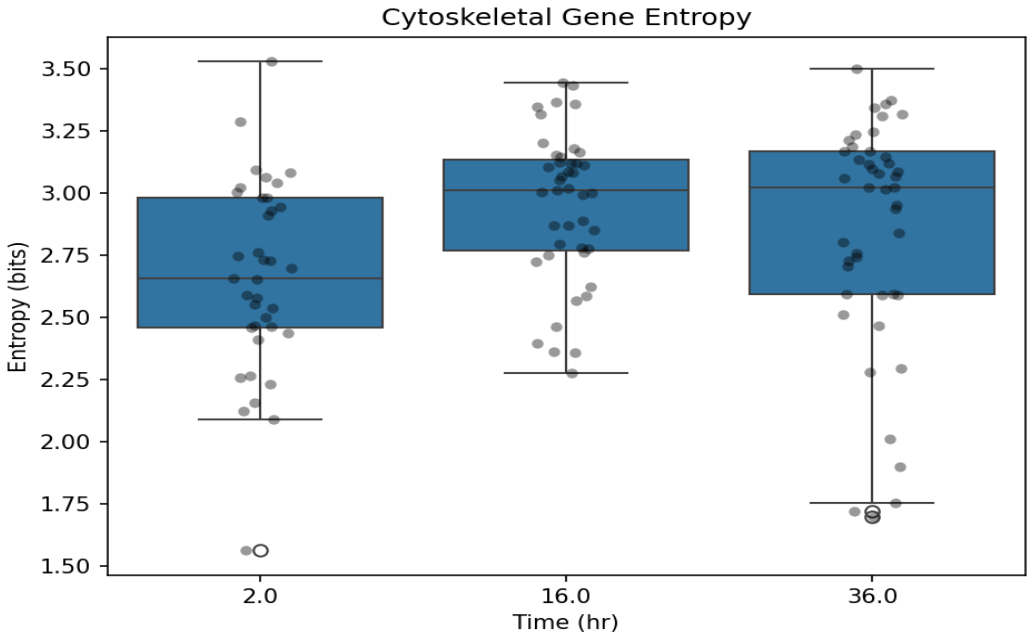
Entropy was computed per cell after converting log2■normalized data to linear scale. Cells were grouped by timepoint. Differential expression was computed for high vs low entropy cells and early vs late timepoints using Welch's t■test. GO enrichment was performed using g:Profiler. Thresholds: $p < 0.05$, $|\log2FC| > 1.0$.

Global Entropy Across Time

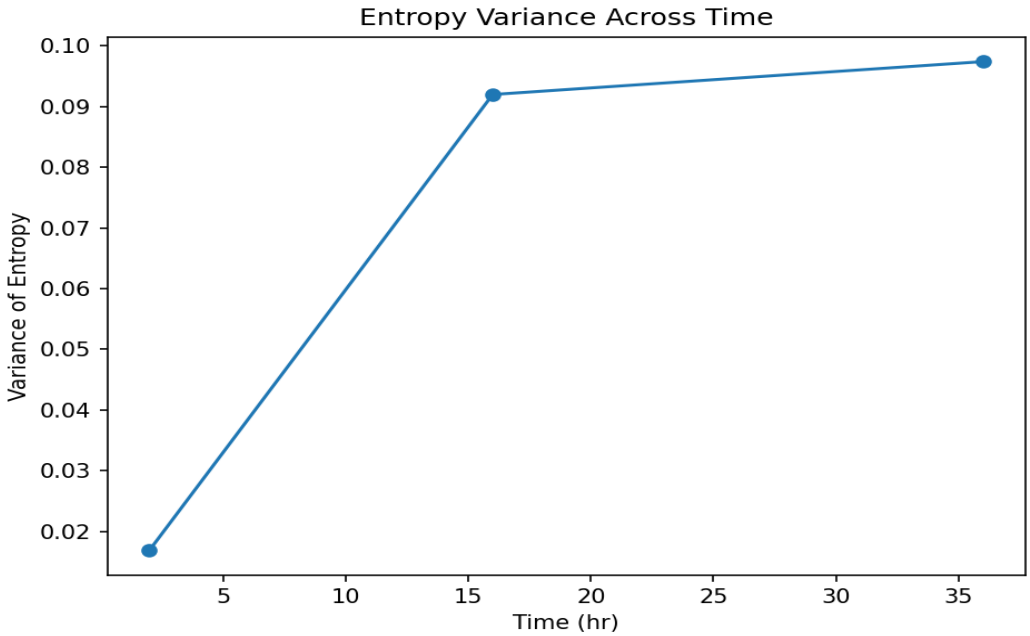
ANOVA p■value: 5.042e-02
Kruskal–Wallis p■value: 1.838e-03



Cytoskeletal Gene Entropy

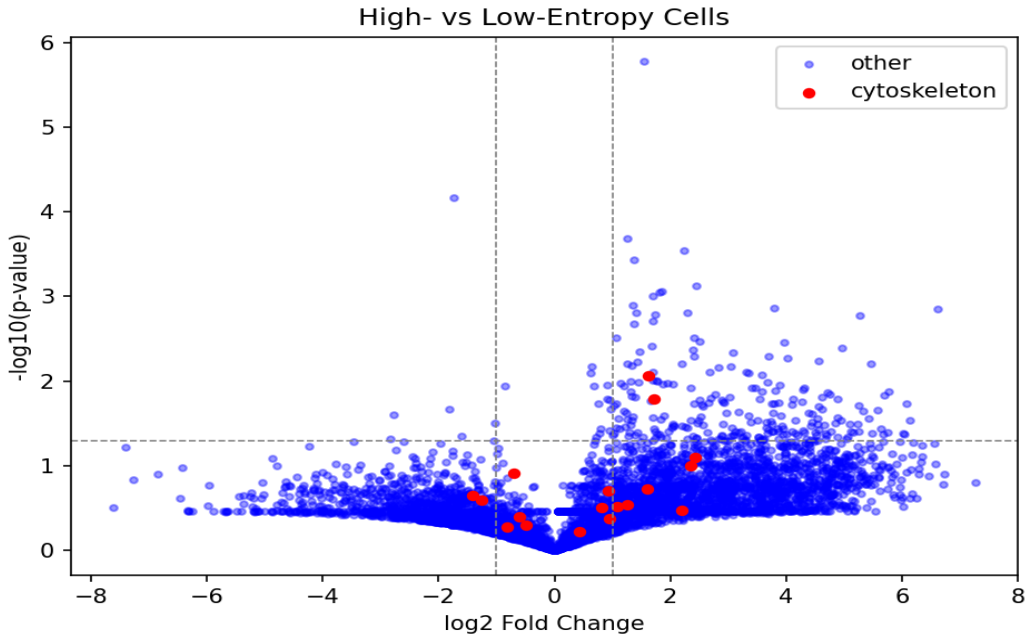


Variance of Entropy Over Time



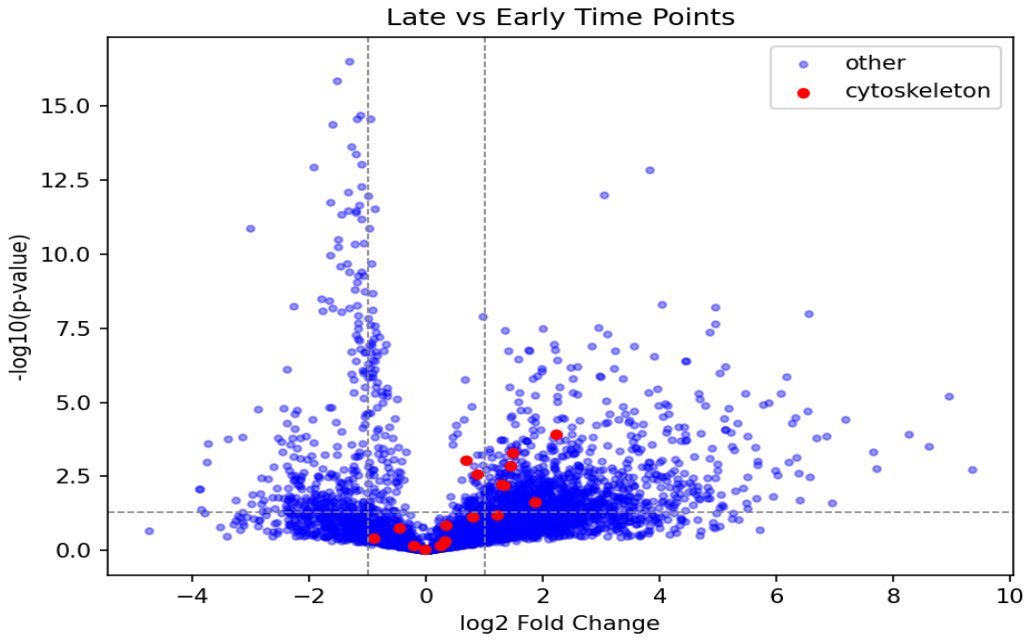
Differential Expression

High■ vs Low■Entropy Cells



Gene	log2FC	p■value
RPS8A	1.5529378204850923	1.68e-06
APC2	-1.7457723859229635	6.86e-05
RPS31	1.2612965351357803	2.04e-04
RPL21B	2.2378227472665118	2.89e-04
RPS23B	1.3834125798458345	3.70e-04
RPS8B	2.4545521203991045	7.41e-04
RPS22A	1.8583565099528376	8.77e-04
RPS5	1.8079975869741405	9.02e-04
RPL14B	1.692047418927242	9.98e-04
RPL29	1.3620021669884292	1.26e-03

Late vs Early Time Points



Gene	log2FC	p■value
MTC7	-1.3223091087442838	3.12e-17
RPL7A	-1.5325832460408442	1.40e-16
ENO2	-1.1229860322537468	1.97e-15
TEF1	-0.9658963745580665	2.65e-15
RPS2	-1.193281080068008	2.72e-15
RPL12A	-1.6018680513337766	4.17e-15
RPL4A	-1.2778898053273393	2.39e-14
RPL18A	-1.2042622761774318	4.02e-14
RPS3	-1.1159440504156768	8.81e-14
RPS0B	-1.9361497342785658	1.16e-13

GO Enrichment (High■ vs Low■Entropy Genes)

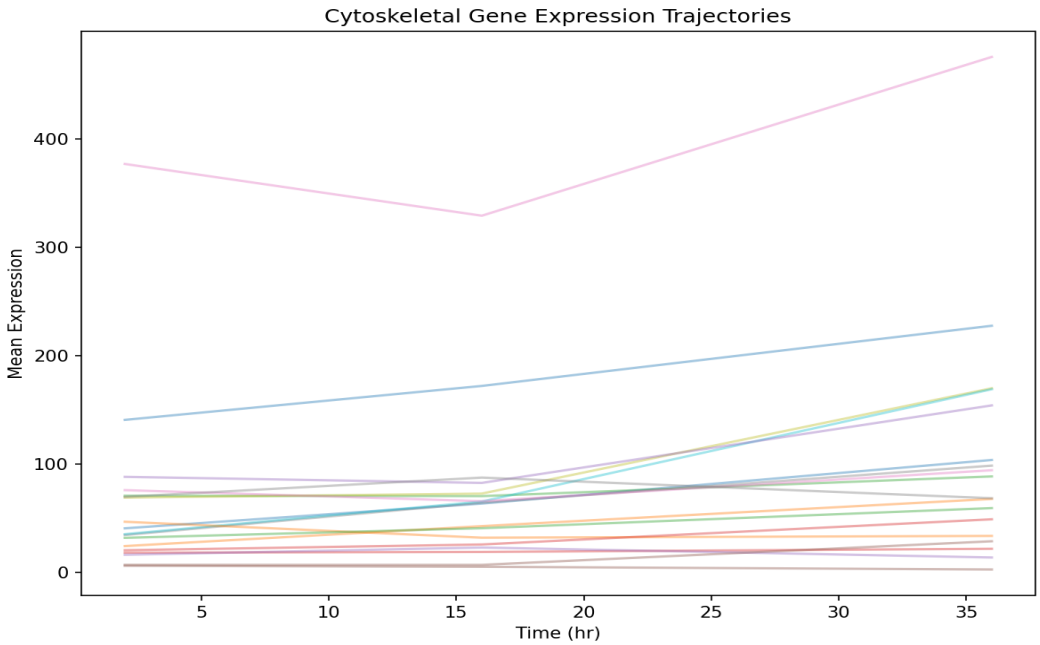
GO term	Description	p■value
GO:0022626	cytosolic ribosome	1.06e-20
GO:0003735	structural constituent of ribosome	2.49e-20
GO:0005840	ribosome	1.02e-19
GO:0002181	cytoplasmic translation	1.31e-19
GO:0044391	ribosomal subunit	9.99e-19

GO:0005198	structural molecule activity	2.40e-16
KEGG:03010	Ribosome	2.94e-16
GO:0006412	translation	3.85e-15
GO:0022627	cytosolic small ribosomal subunit	7.70e-14
GO:1990904	ribonucleoprotein complex	1.46e-12

GO Enrichment (Late vs Early Genes)

GO term	Description	p■value
GO:0022626	cytosolic ribosome	2.33e-50
GO:0005840	ribosome	5.35e-47
GO:0002181	cytoplasmic translation	3.34e-46
GO:0003735	structural constituent of ribosome	1.36e-44
GO:0044391	ribosomal subunit	4.27e-44
GO:0005198	structural molecule activity	1.37e-37
GO:0006412	translation	1.85e-35
KEGG:03010	Ribosome	4.14e-35
GO:1990904	ribonucleoprotein complex	4.52e-31
GO:0005829	cytosol	6.22e-25

Cytoskeleton Gene Expression Over Time



Gene Classification

Core aging (both lists): 7
Entropy■specific: 43
Time■specific: 43
Background: 5904

Overlap of top 50 genes: 7 genes

Genes in overlap: YLR167W, YPR132W, YGL123W, YJL190C, YPR102C, YML026C, YJR123W